

Seismic imaging reveals ongoing modification of craton margins in Northeast Asia

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Figures S1 to S18

Introduction

This Supplementary Information provides additional data, figures, and methodological details supporting the results presented in the main manuscript.

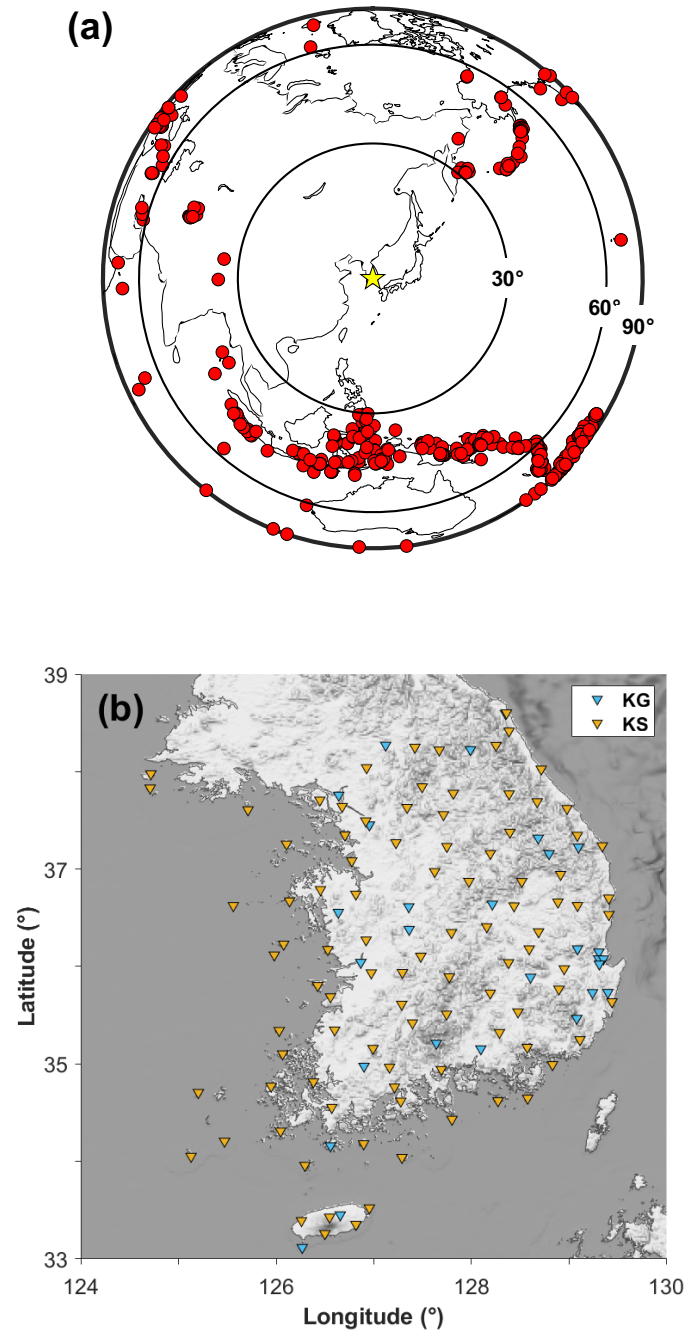


Figure S1. (a) Distribution of teleseismic events (red circles) at epicentral distances of 30°–90° from the reference location ([36°N, 127°E], yellow star) used in this study. (b) Distribution of broadband seismic stations within the study region. Inverted triangles indicate KG (blue) and KS (yellow) stations operated by the Korea Institute of Geoscience and Mineral Resources (KIGAM) and the Korea Meteorological Administration (KMA), respectively.

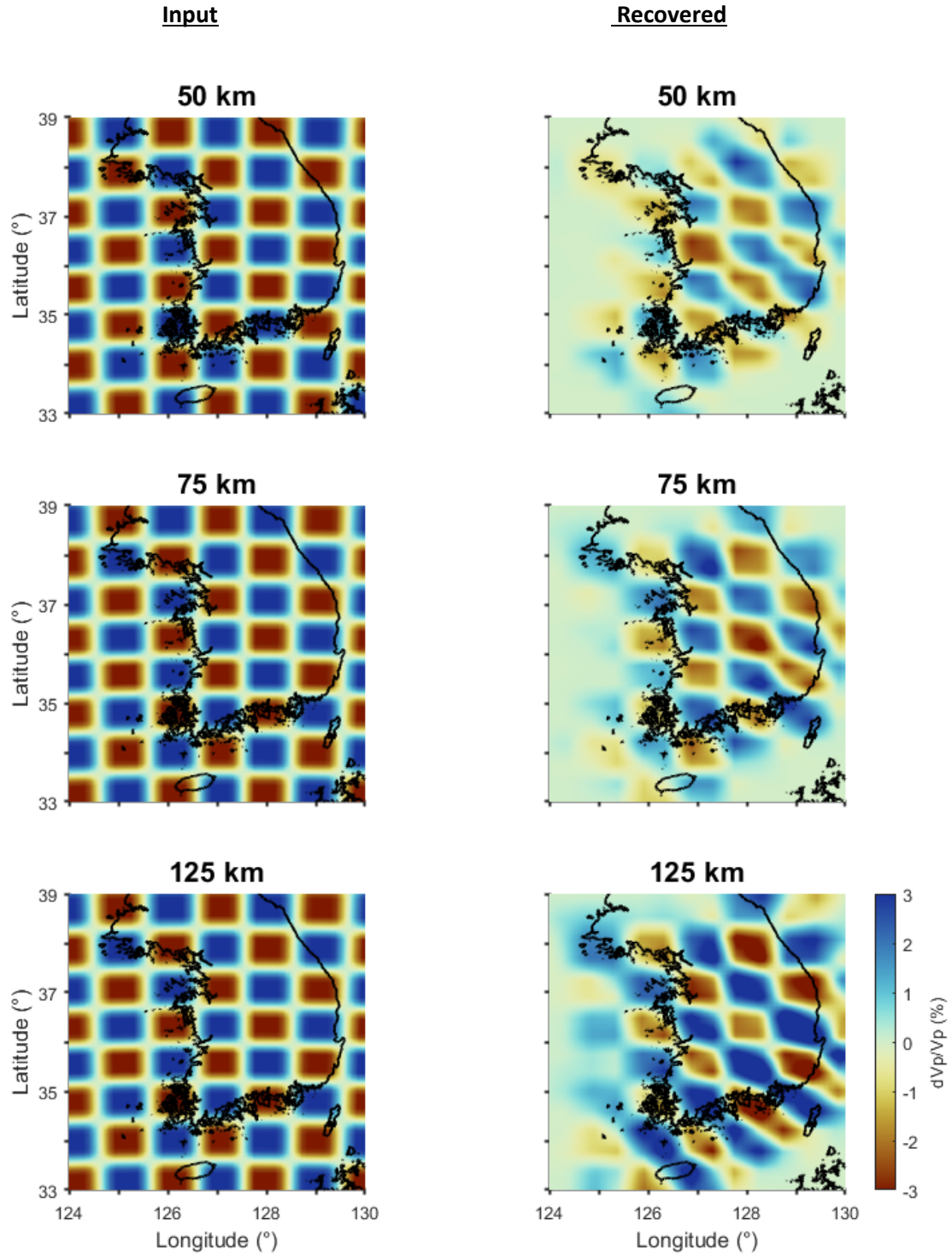


Figure S2. Checkerboard test results demonstrating a lateral resolution of $50 \times 50 \text{ km}^2$ for our tomography model. The left column figures show the input checkerboard velocity anomalies, while the right column figures show the recovered anomalies.

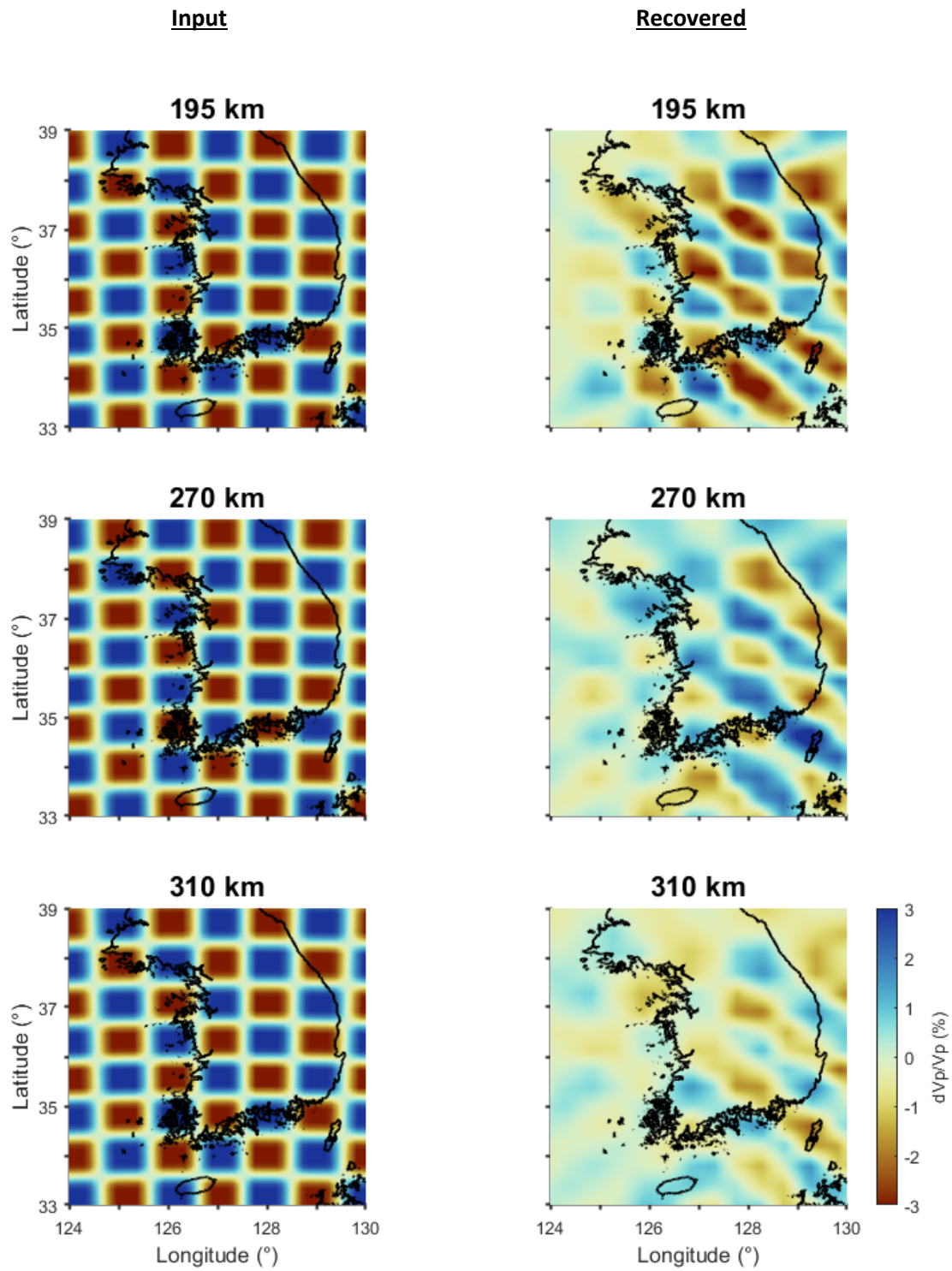


Figure S2. Cont'd.

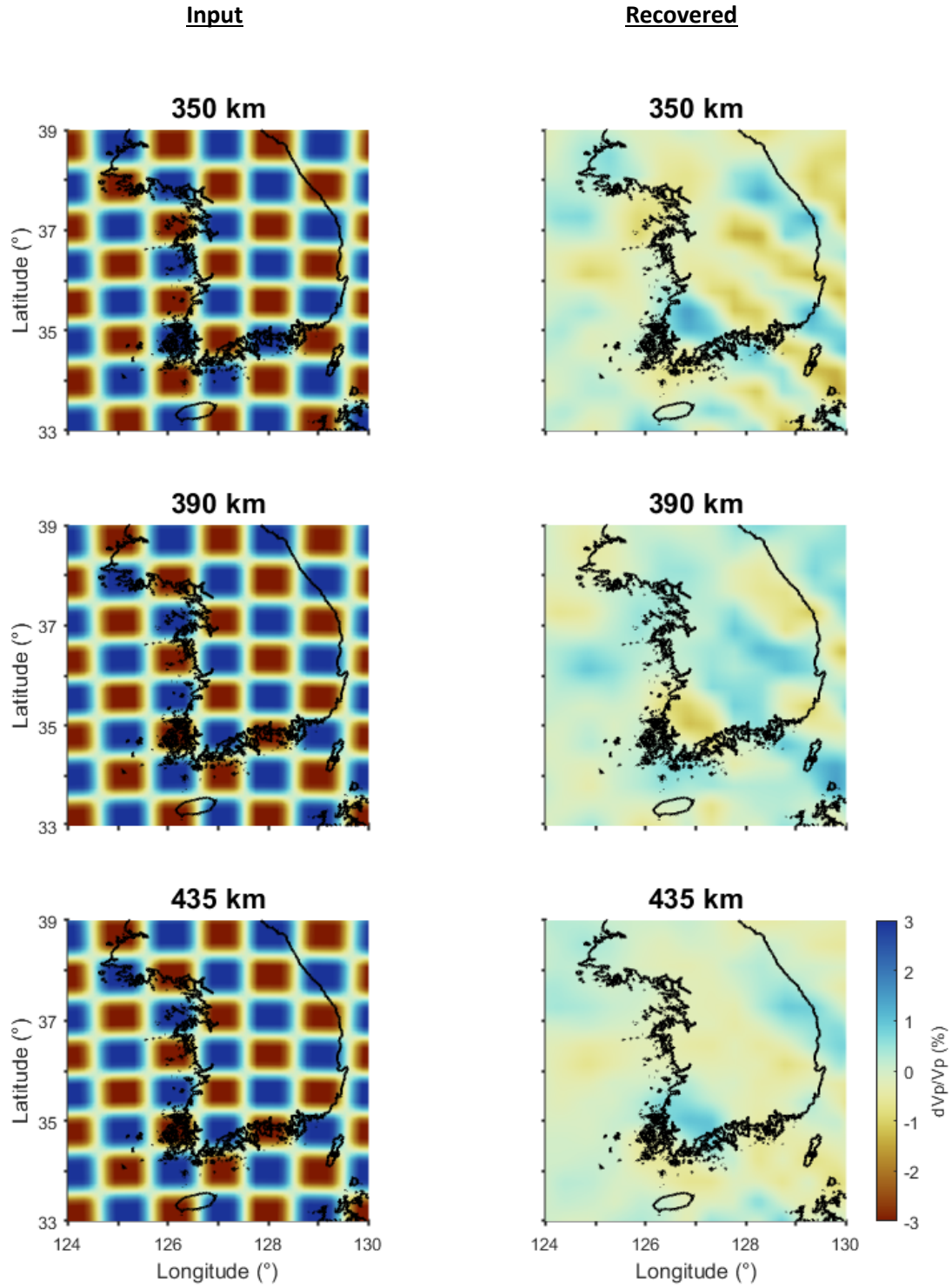


Figure S2. Cont'd.

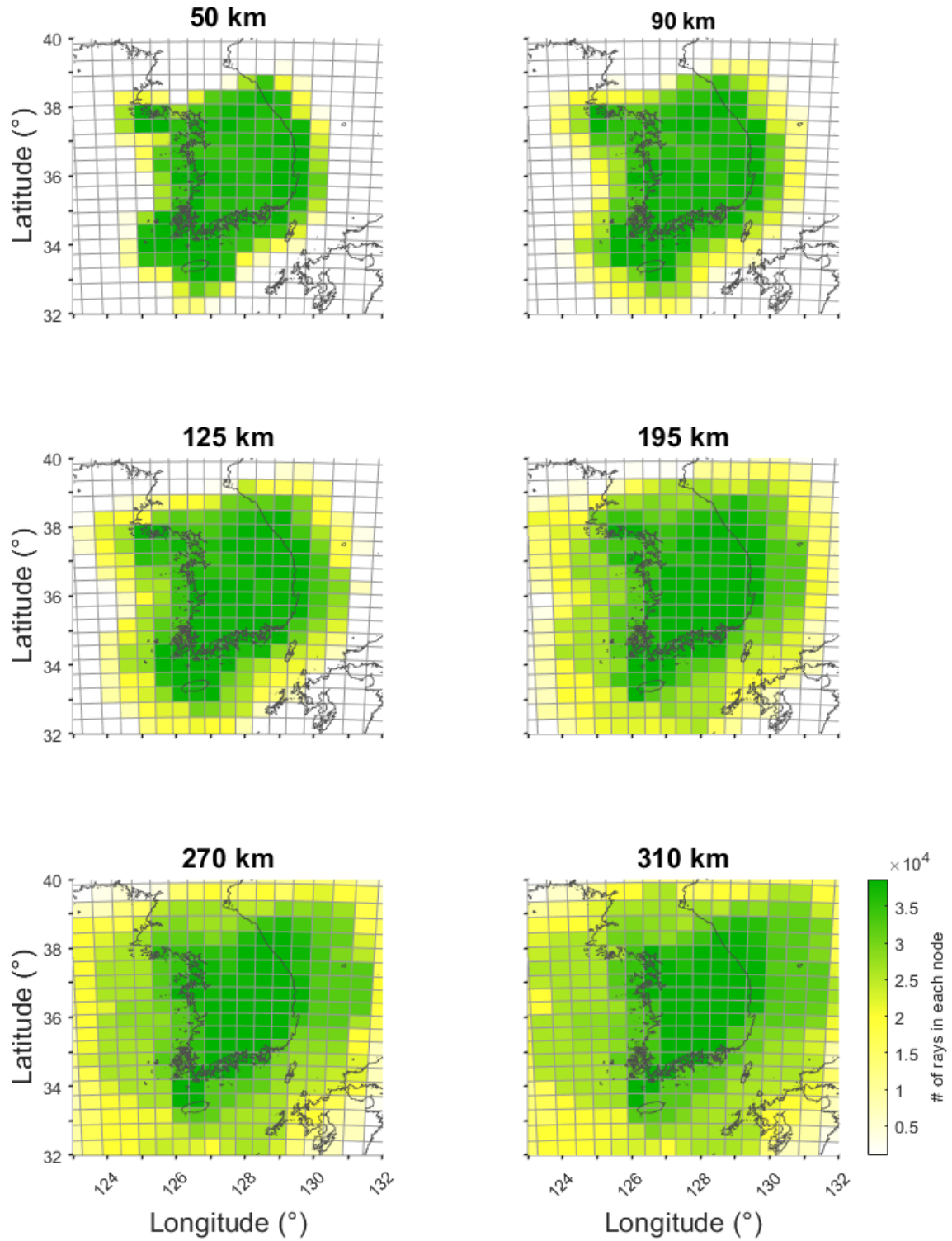


Figure S3. Results of the hit quality test for seismic ray coverage of the study area.

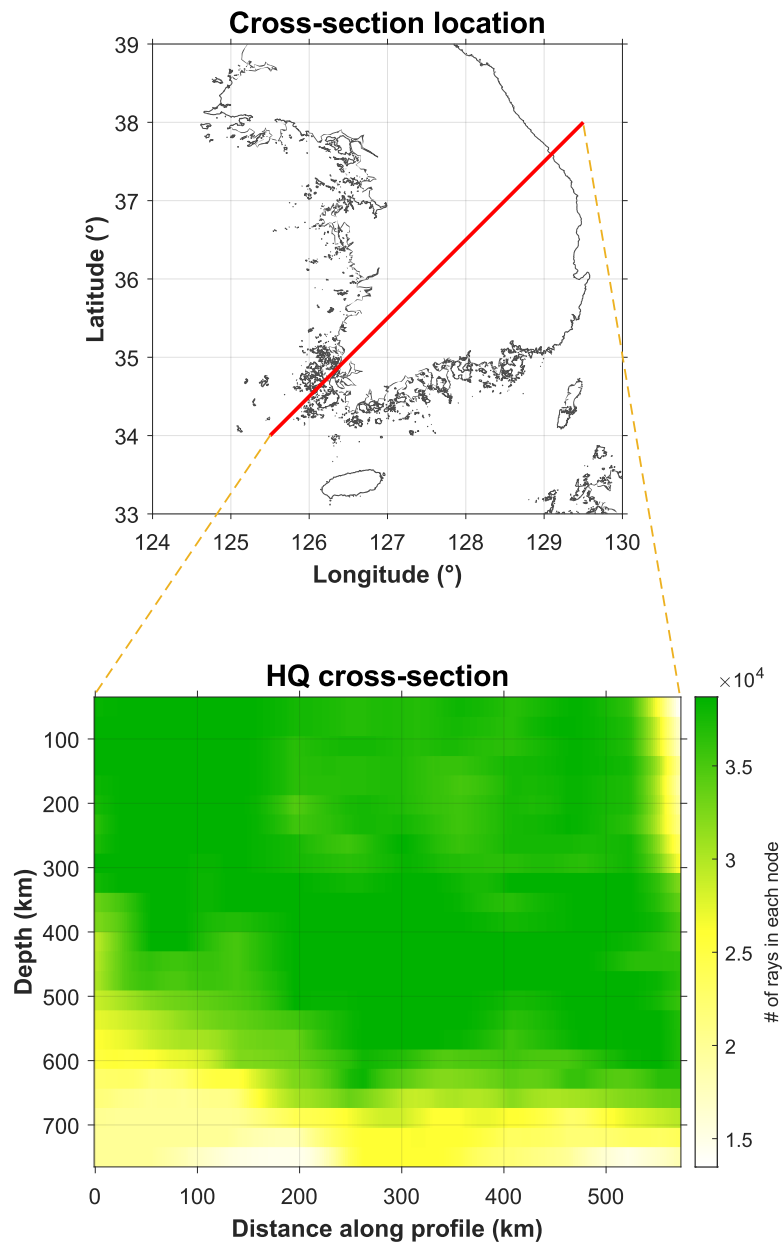


Figure S3. Cont'd. The red solid line in the top panel indicates the profile location of the cross section shown below.

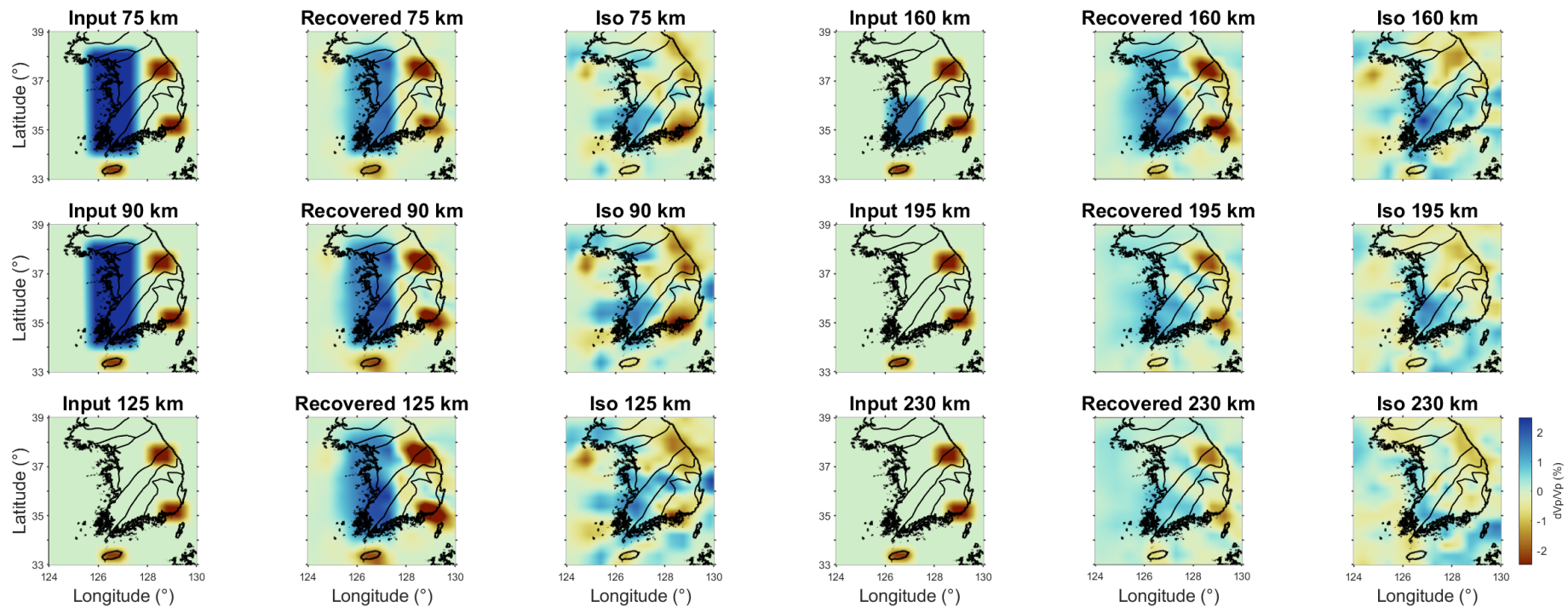


Figure S4. Examples of synthetic tests designed to evaluate lithospheric delamination, shown as depth slices by depth. Panels compare the input model, the recovered model, and results from isotropic inversion.

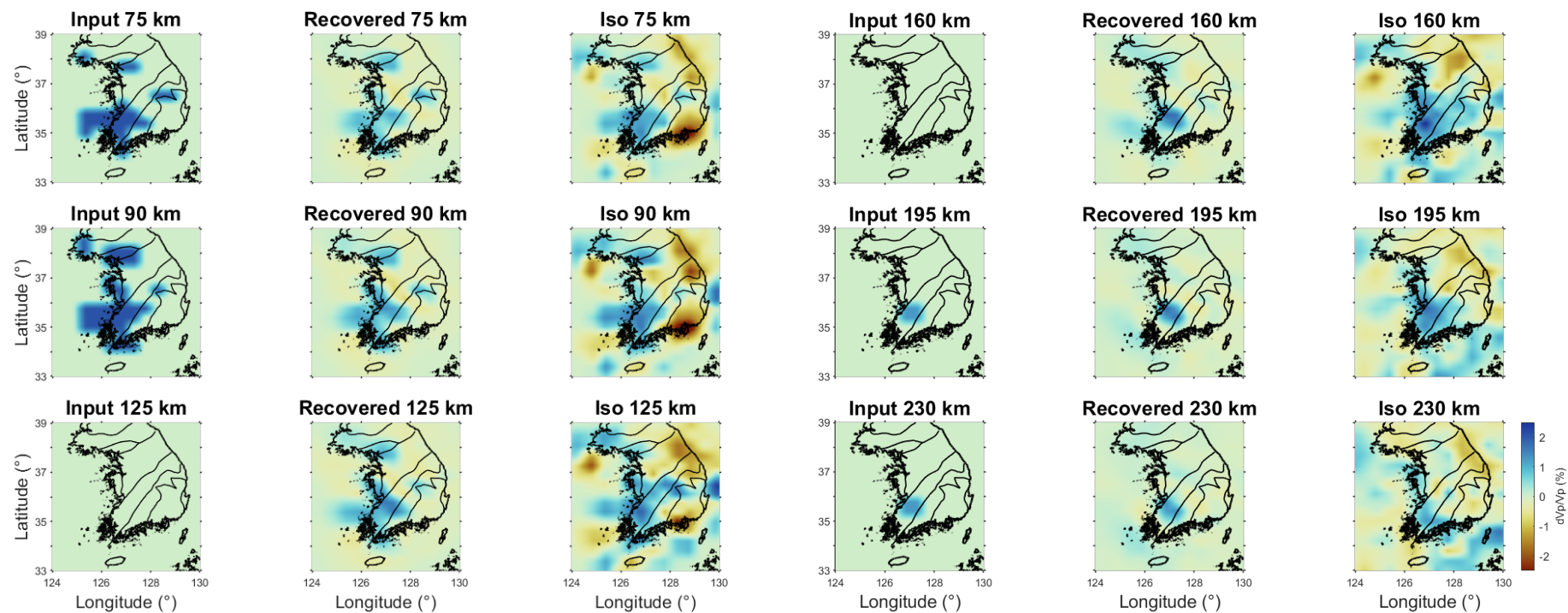


Figure S4. Cont'd

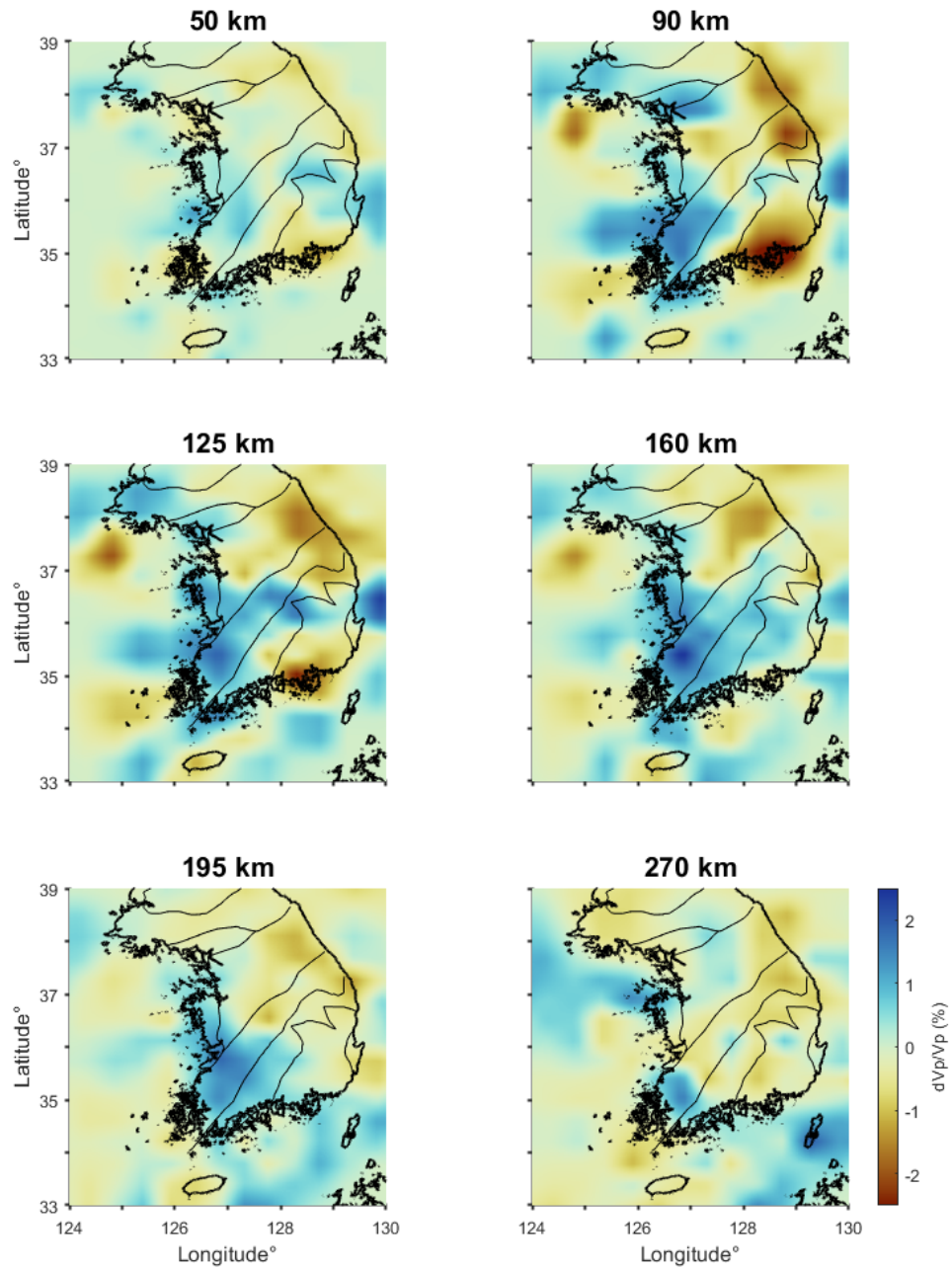
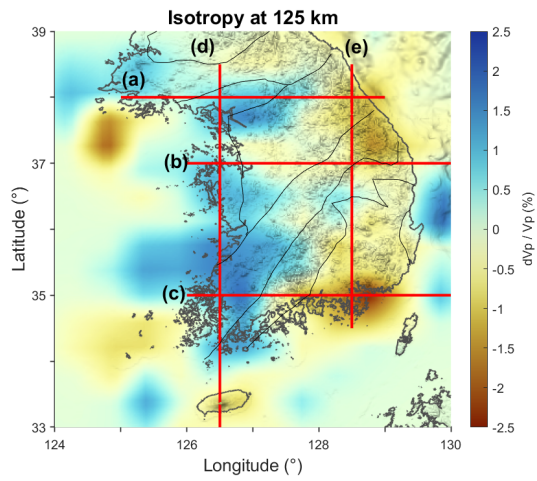


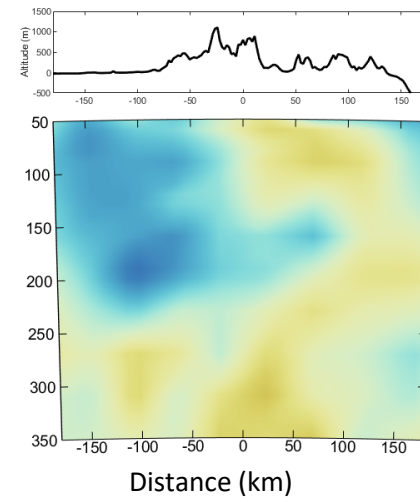
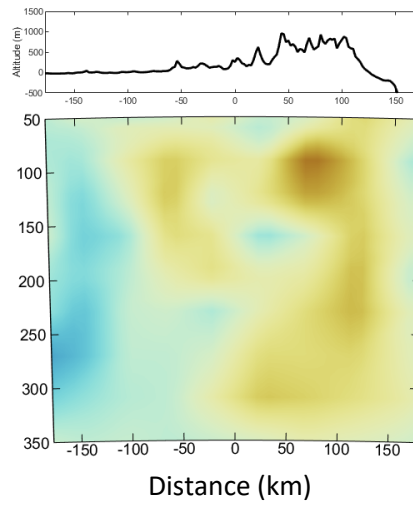
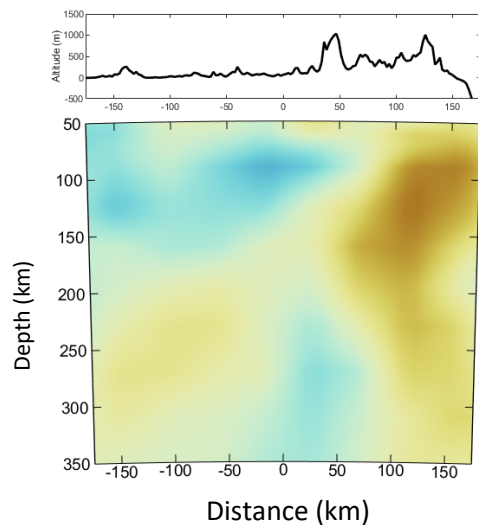
Figure S5. Depth slice images of isotropic P-wave tomography at 50, 90, 125, 160, 195 and 270 km depth.



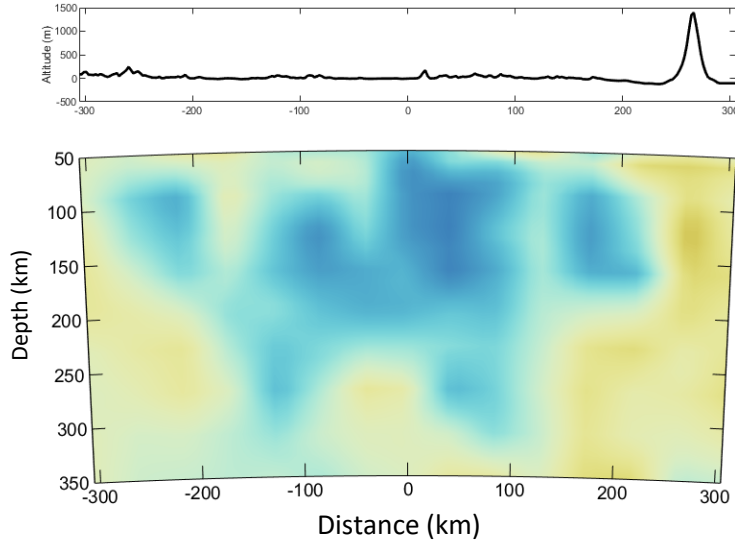
(a)

(b)

(c)



(d)



(e)

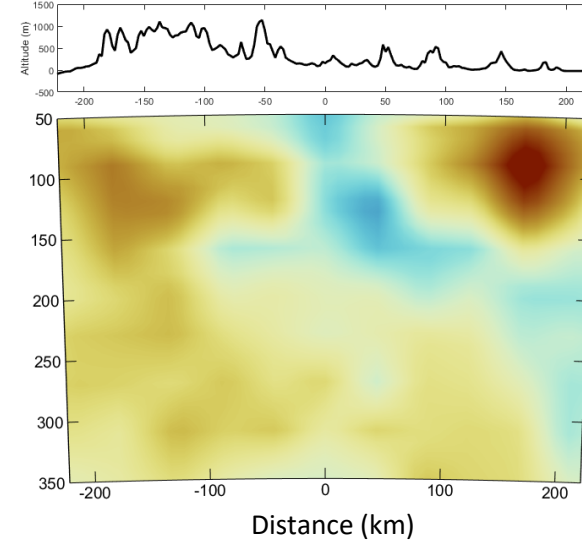


Figure S6. Cross-sectional views of the isotropic tomography model along the red solid lines shown in the first subfigure. The cross sections correspond to the following profiles: **(a)** [38°N, 125°E] to [38°N, 129°E], **(b)** [37°N, 126°E] to [37°N, 130°E], **(c)** [35°N, 126°E] to [35°N, 130°E], **(d)** [38.5°N, 126.5°E] to [33°N, 126.5°E], and **(e)** [38.5°N, 128.5°E] to [34.5°N, 128.5°E]. These cross sections reveal low-velocity anomalies beneath the mountainous eastern Korean Peninsula, in contrast to high-velocity anomalies beneath the flat western region.

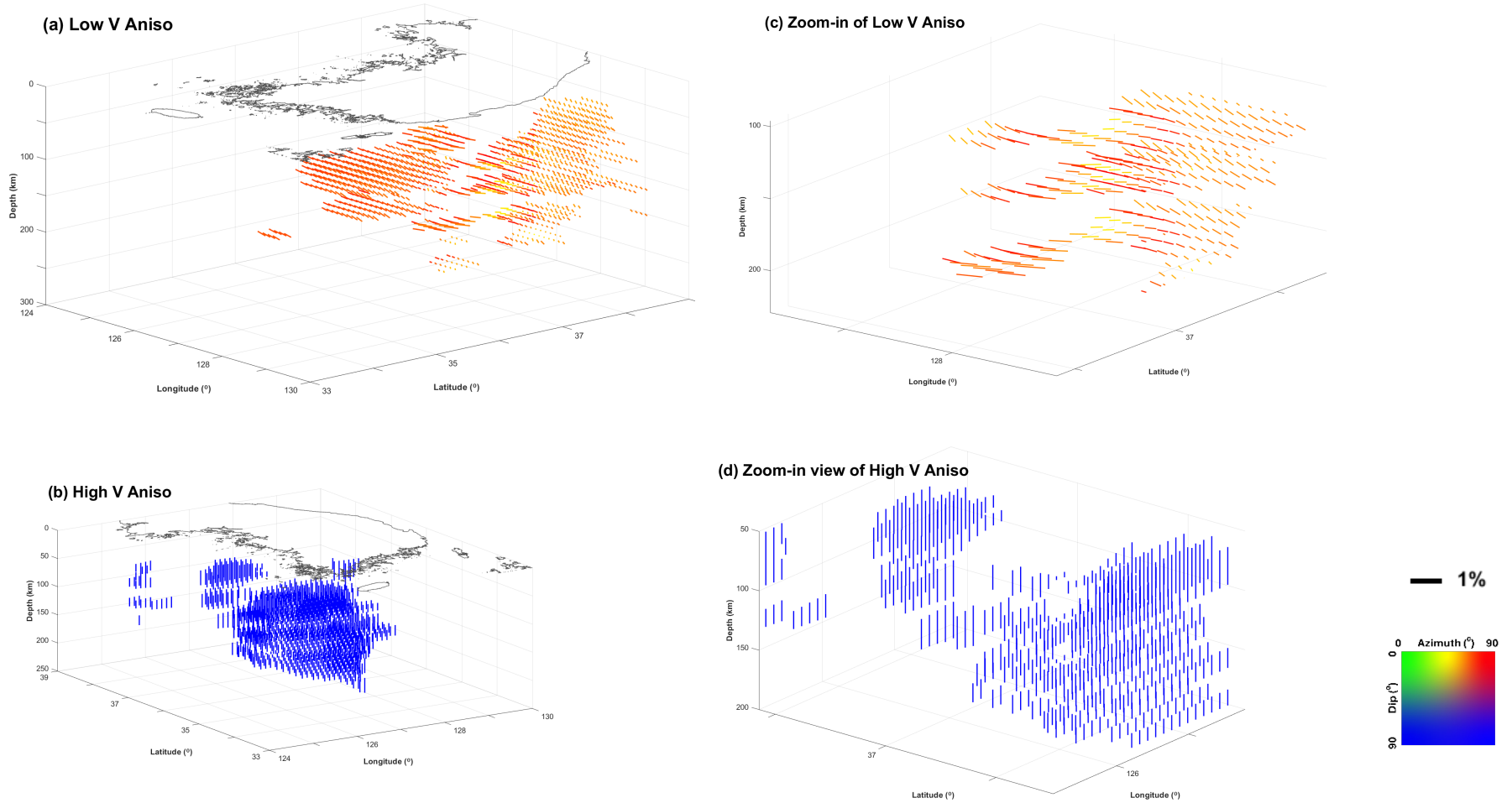


Figure S7. Anisotropy models for **(a)** low-velocity anomalies (LVAs) in the eastern Korean Peninsula (KP), and **(b)** high-velocity anomalies (HVAs) in the western KP. **(c)** Zoomed-in view of the anisotropy model highlighting the LVAs shown in (a), focused on the region between 127–129°E longitude and 36–38°N latitude at depths of 100–200 km. **(d)** Zoomed-in view of the anisotropy model highlighting the HVAs shown in (b), focused on the region between 125–127°E longitude and 34–36°N latitude at depths of 100–200 km.

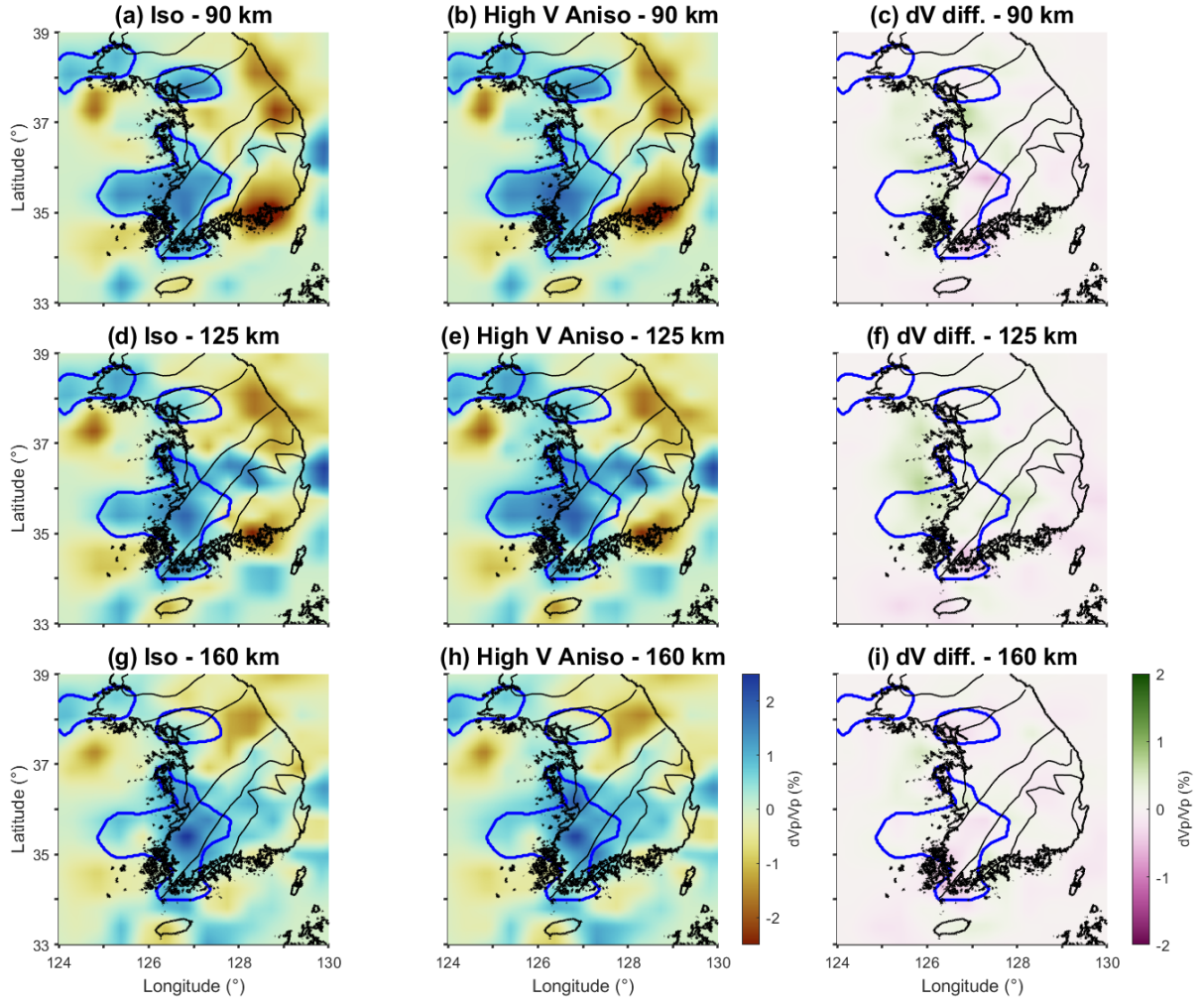


Figure S8. Comparison of isotropic and anisotropic models highlighting the HVA in the western KP. The blue contour lines indicate the region where the prescribed anisotropy model, High V Aniso, is applied. **(a)**, **(d)**, and **(g)** show the tomographic results of the isotropic inversion at depths of 90, 125, and 160 km respectively. **(b)**, **(e)**, and **(h)** show the tomographic results of the prescribed anisotropy model at depths of 90, 125, and 160 km respectively. The insignificant changes in dV_p/V_p in **(c)**, **(f)**, and **(i)** indicate that the prescribed anisotropy model has little impact on the HVAs relative to the isotropic tomography results at 90, 125, and 160 km, respectively. Black solid lines within the KP indicate major geological boundaries.

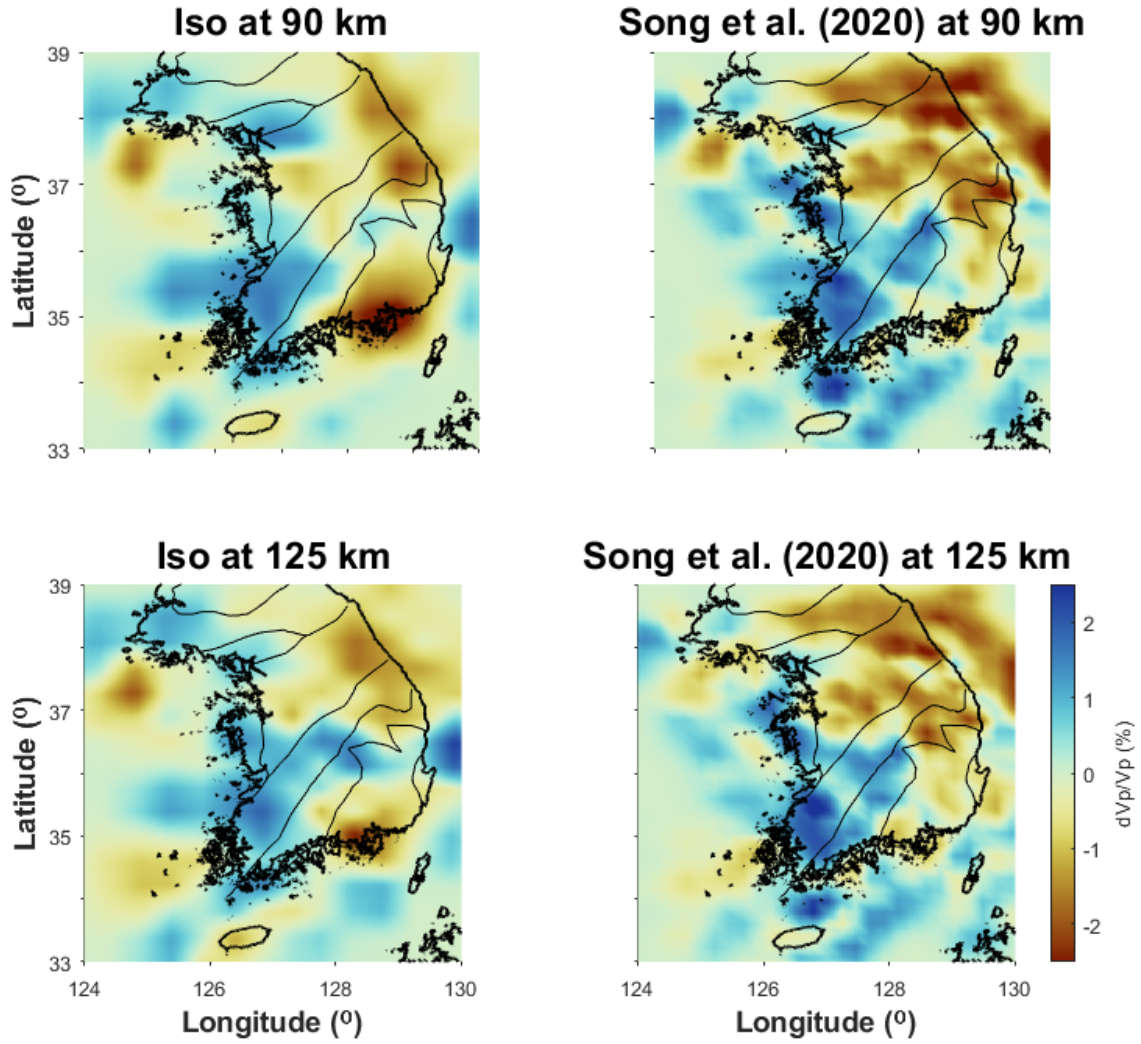


Figure S9. Side-by-side comparison of isotropic P-wave models at 90 and 125 km depth: (left) the model from this study and (right) that of the previous study (Song et al., 2020).

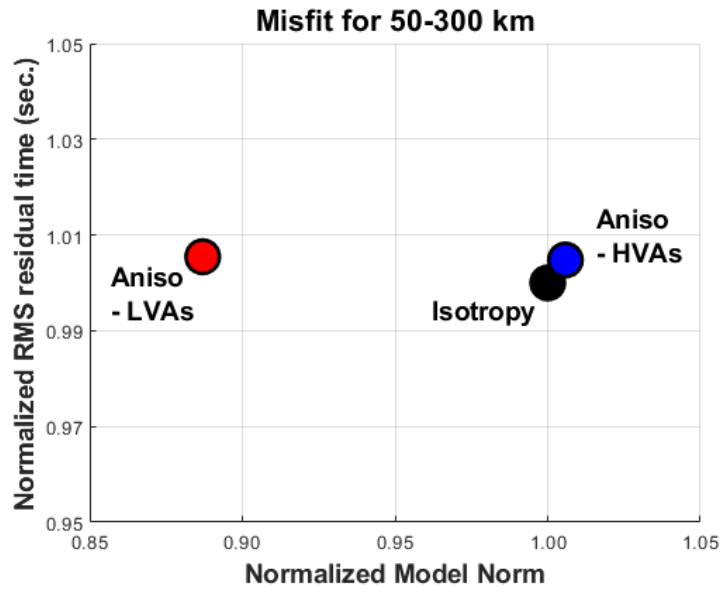


Figure S10. Root-mean-square (RMS) travel-time residuals and normalized L2 model norm misfits (normalized to the isotropic model) for the isotropic and two anisotropic models over the 50–300 km depth range.

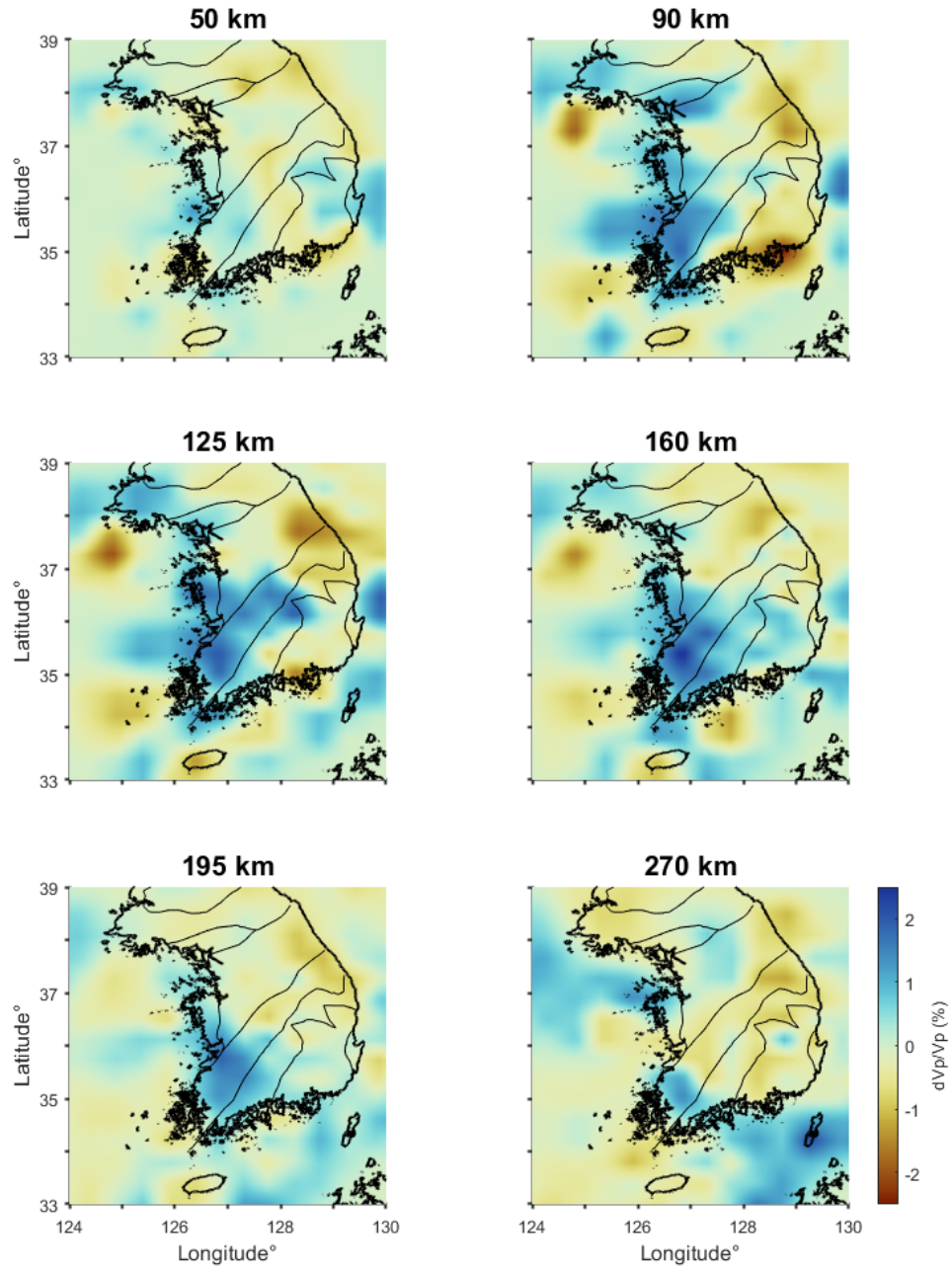


Figure S11. Anisotropic tomography depth slices incorporating anisotropy in the eastern low-velocity anomaly (LVAs) region, as outlined by the red contours in Figure 1.

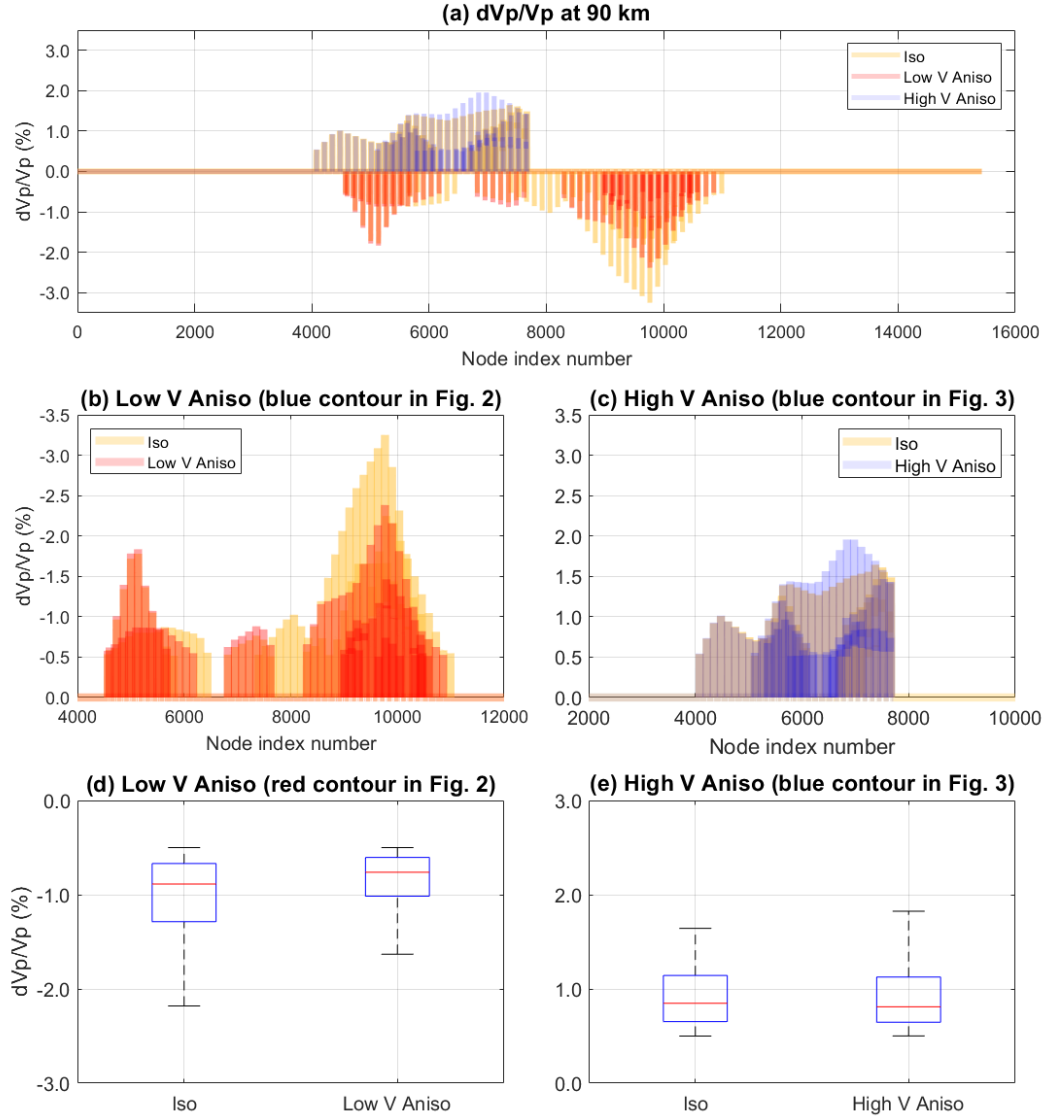


Figure S12. (a)–(c) show bar graphs of the dV_p/V_p at each node at a depth of 90 km. Only nodes located within the red and blue contour lines in Figure 1 and Figure S7, respectively, are shown. Specifically, (b) shows dV_p/V_p values within the red contour lines in Figure 1 (i.e., LVAs in the eastern KP), while (c) shows dV_p/V_p values within the blue contour lines in Figure S7 (i.e., HVAs in the western KP). (d) and (e) display boxplots of dV_p/V_p for the same regions shown in (b) and (c), respectively. (b) and (d) show the reduction in the target low-velocity anomalies in the anisotropic tomography results incorporating the anisotropy model, Low V Aniso; while (c) and (e) show no significant change in the target high-velocity anomalies in the anisotropic tomography results incorporating the anisotropy model, High V Aniso.

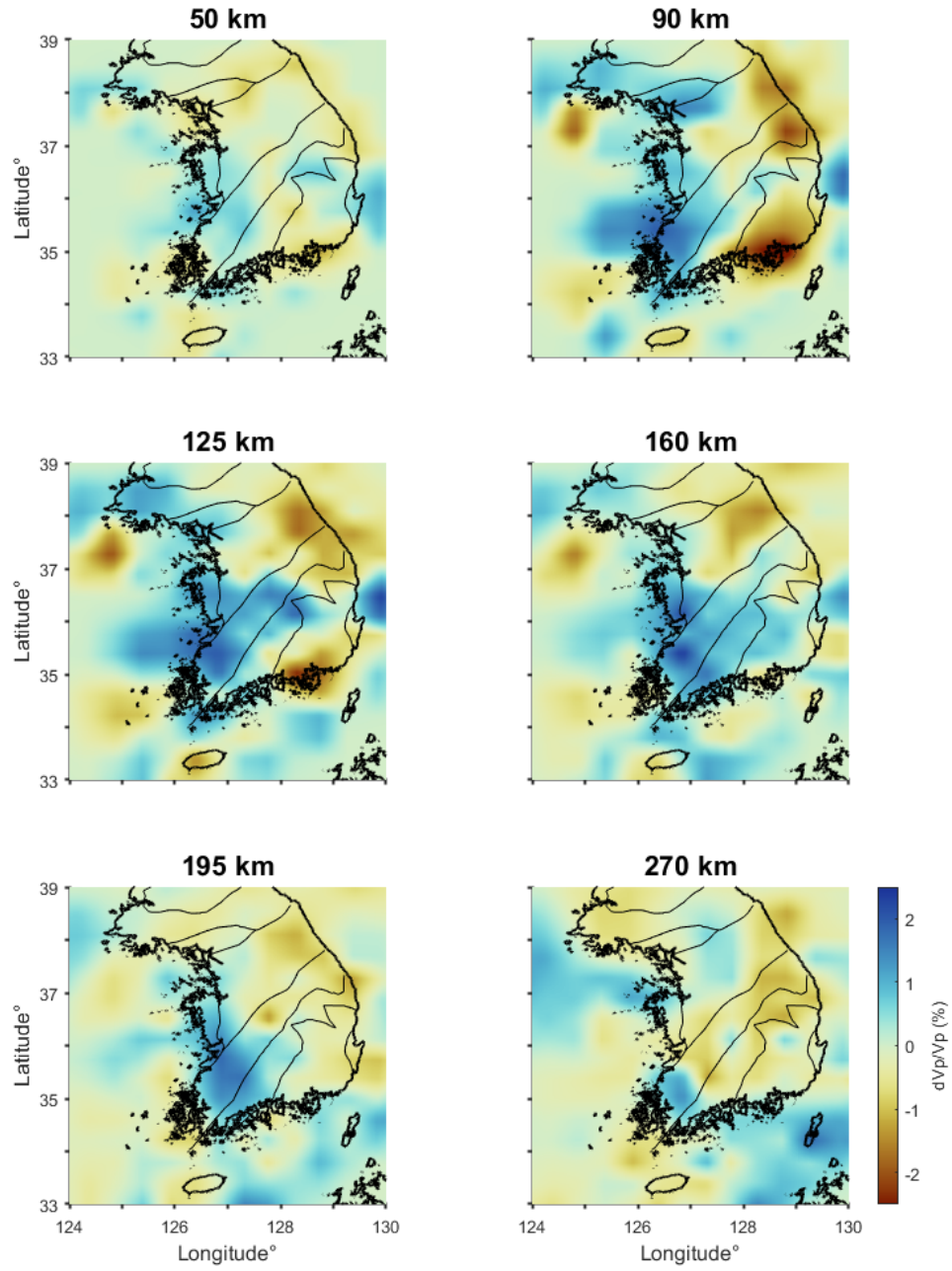


Figure S13. Anisotropic tomography depth slices incorporating anisotropy in the western high-velocity anomaly (HVAs) region, as outlined by the blue contours in Figure S7.

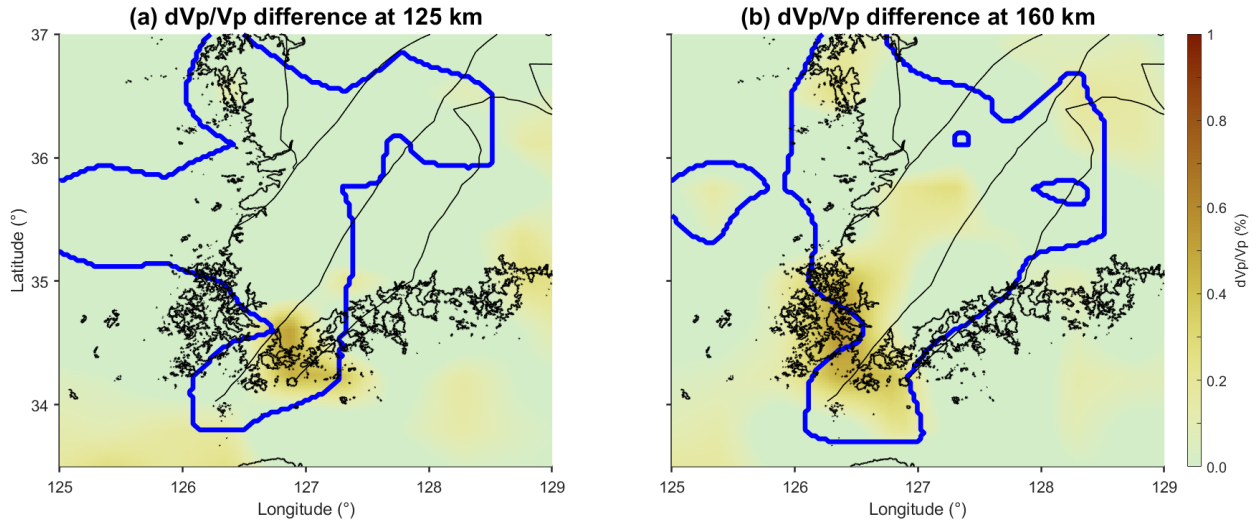


Figure S14. Difference in dV_p/V_p between isotropic and anisotropic velocity models for the HVAs in the western KP (i.e., isotropic minus anisotropic dV_p/V_p for 34.5–37 °N latitude, 125–129 °E longitude). The blue solid contour line indicates lateral extent of the HVA anisotropy model at depths of **(a)** 125 km and **(b)** 160 km. We find a minor reduction ($\sim 0.7\%$) at 125 and 160 km depth when the anisotropy model is incorporated, which features a vertically oriented seismically fast direction (SFD) representing vertical mantle flow. This may represent the presence of downwelling mantle flow in the western KP.

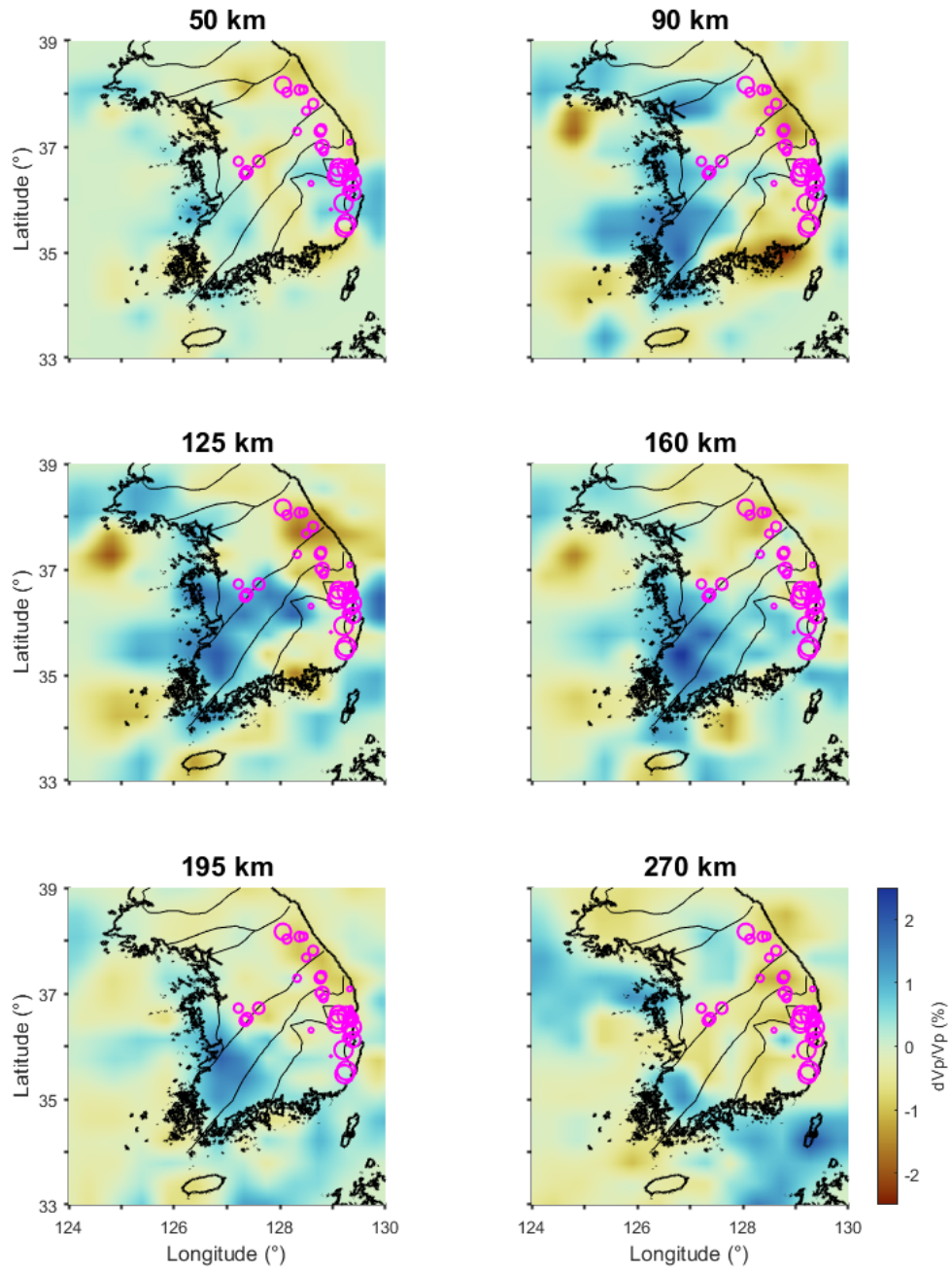


Figure S15. Distribution of observed mantle-sourced helium isotopes. It is corrected helium isotope ratio ($^3\text{He}/^4\text{He}$) expressed as R_c/R_a (magenta circle) (R_c : corrected $^3\text{He}/^4\text{He}$ for atmospheric contamination; R_a : the atmospheric $^3\text{He}/^4\text{He}$ as a reference value). All data plotted show $R_c/R_a > 1$, indicating a mantle-derived helium source (Kim et al., 2024). The size of each circle is proportional to the magnitude of the R_c/R_a ratio.

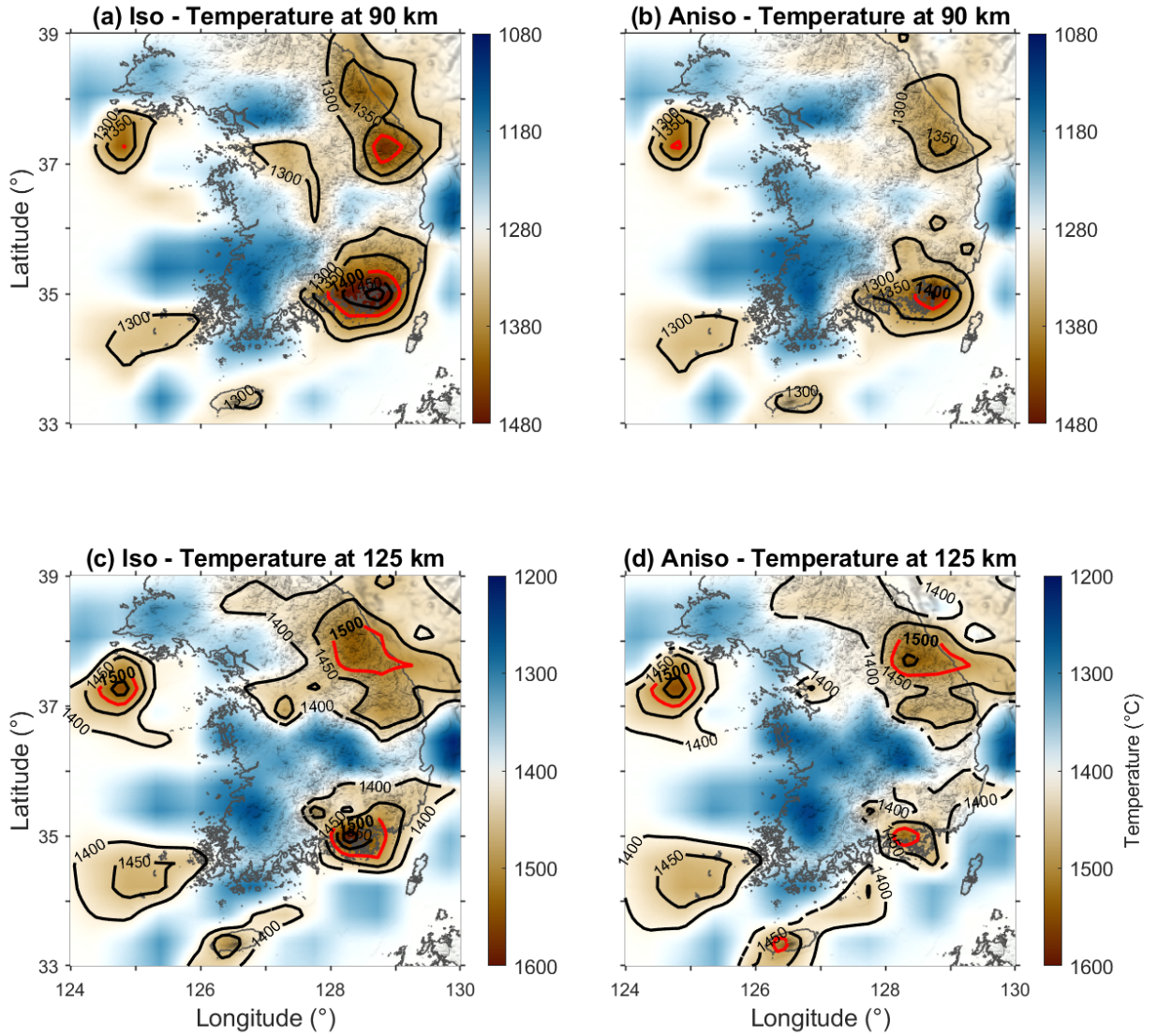


Figure S16. Temperature estimations at 90 km depth based on (a) isotropic velocity model and (b) prescribed anisotropy velocity model targeting LVAs. Temperature estimations at 125 km depth based on (c) isotropic velocity model, and (d) prescribed anisotropy velocity model targeting LVAs. The estimation is based on converting dV_P/V_P values to temperatures at each depth. Pressures and ambient mantle temperature (T) at 90 km (a and b) and 125 km (c and d) are assumed to be 3 and 4 GPa, and 1250 and 1370 °C, respectively. The red solid contour represents the dry solidus temperature of peridotite at each depth (1400 °C in a and b) and 125 km (1500 °C in c and d). Including the anisotropy model for the LVAs in the eastern KP reduces the extent of melt presence in the mantle (b and d). The empirical equations that we used to derive the mantle temperature at 90 and 125 km slices are:

$$dT_{P_{90km}} \cong (-85.5 * dV_P) + 1250 \quad \& \quad dT_{P_{125km}} \cong (-84.7 * dV_P) + 1370$$

based on Figure 3(a) of Cammarano et al. (2003).

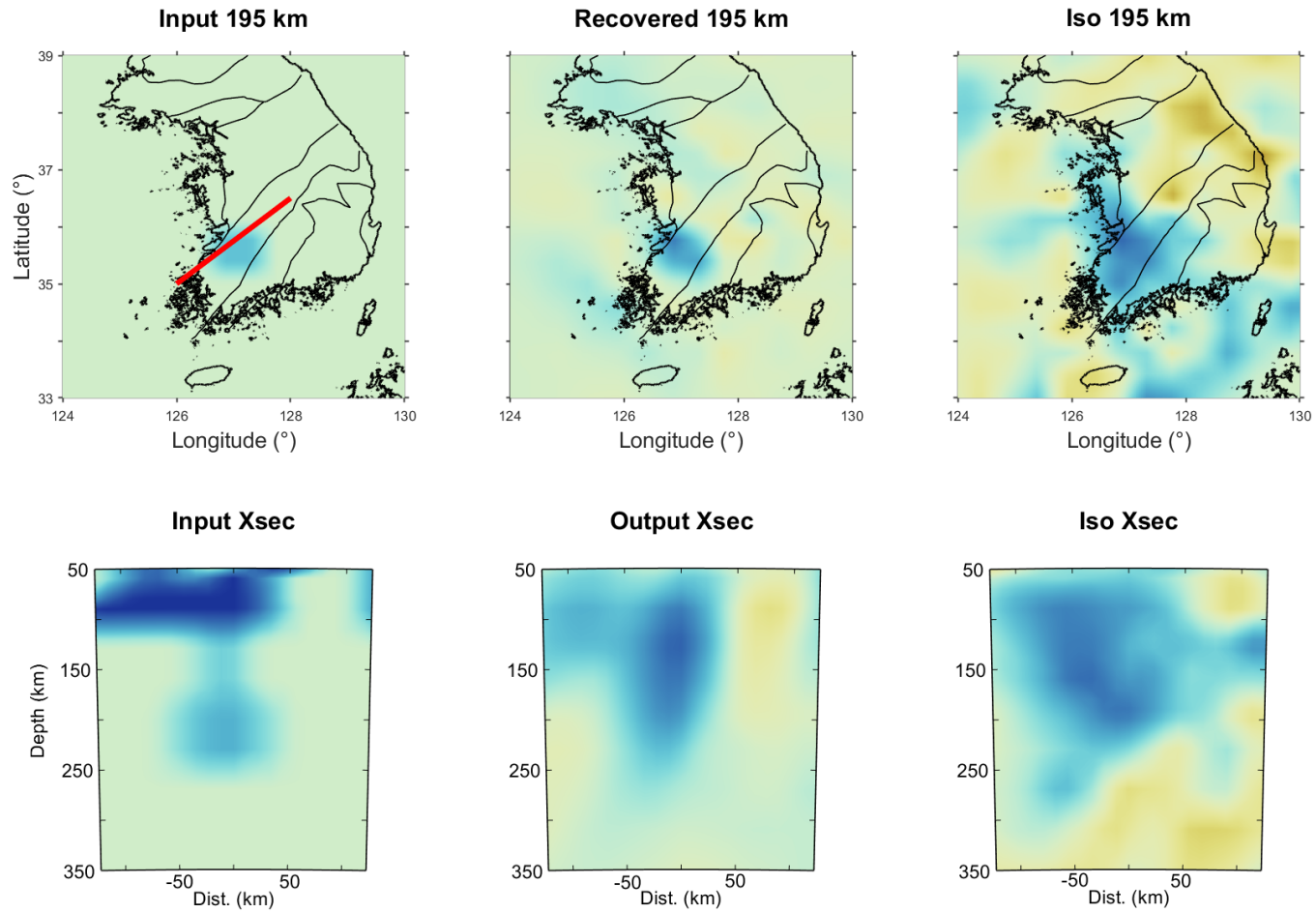


Figure S17. An example of synthetic tests designed to evaluate hanging lithosphere. Panels compare the input model, the recovered model, and results from isotropic inversion. The red solid line in the first panel indicates the location of cross sections.

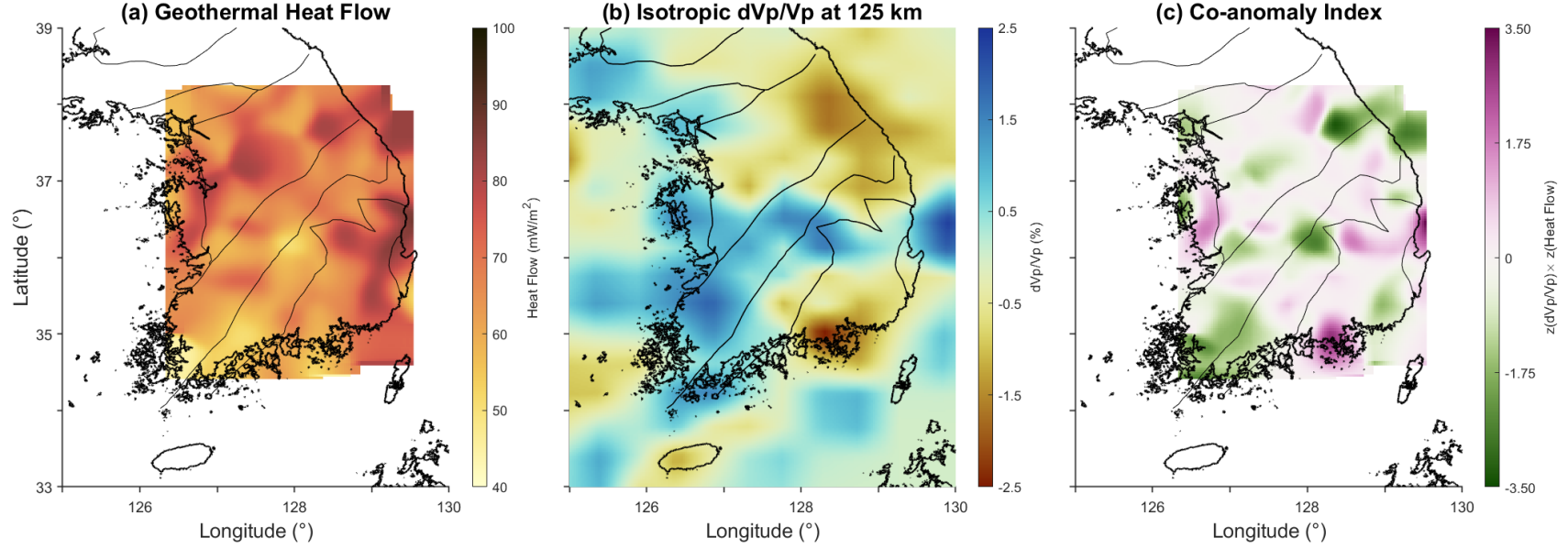


Figure S18. Comparison between **(a)** geothermal heat flow (Kim & Lee, 2007), **(b)** isotropic dV_p/V_p at 125 km, and **(c)** the co-anomaly index showing the correlation between **(a)** and **(b)** based on Z-score analysis. As geothermal heat and dV_p/V_p are negatively correlated, regions exhibiting negative correlation are colored green, whereas regions where geothermal heat and dV_p/V_p are not negatively correlated (i.e., positively correlated) are colored magenta.

References

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